Instantiating Objects in XAML

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Module Outline

Work with elements and attributes

Set properties in XAML

- Attribute syntax
- Property element syntax
- Content syntax
- Collection syntax

Work with Elements and Attributes

Element

Closing Element

```
<Button Content="Add customer"></Button>
```

Work with Elements and Attributes

Element

Self-closing

```
<Button Content="Add customer"/>
```

Work with Elements and Attributes

Element Is mapped to a **class**

Attribute Is mapped to a **property**

```
<Button Content="Add customer"
        Click="Button_Click"/>
```

Attribute Is mapped to an **event**

Work with Elements and Attributes

```
<Button Content="Add customer"
        Click="Button_Click"/>
```

C#

```
var btn = new Button
{
  Content = "Add customer"
};

btn.Click += Button_Click;
```

Demo

Explore the button in the Customers App

Demo

Set properties with the property element syntax

Set Properties with Content Syntax

Attribute syntax

```
<Button Content="Add customer"/>
```

Property element syntax

```
<Button>
  <Button.Content>
    Add customer
  </Button.Content>
</Button>
```

Content syntax

```
<Button>
  Add customer
</Button>
```

Set Properties with Content Syntax

```
<Button>
  Add customer
</Button>
```

XAML parser

1. **Content without property element**
2. **Looks for the** `ContentPropertyAttribute`
3.
```
[ContentProperty(Name = "Content")]
public class ContentControl { ... }
```
4. **Uses the property specified in the attribute**

Content syntax is always used when you put content in an object element without property element

Demo

Set properties with the content syntax

Use Collection Syntax

Attribute syntax

Property element syntax

Content syntax

Use Collection Syntax

```
<StackPanel>
  <StackPanel.Children>
    <TextBlock/>
  </StackPanel.Children>
</StackPanel>
```

```
public class Panel : FrameworkElement
{
  public UIElementCollection Children { get ... }
  ...
}
```

Use Collection Syntax

```
<StackPanel>
  <StackPanel.Children>
    <TextBlock/>
  </StackPanel.Children>
</StackPanel>
```

XAML parser

```
var stackPanel = new StackPanel();
stackPanel.Children.Add(new TextBlock());
```

Use Collection Syntax

```
<StackPanel>
  <StackPanel.Children>
    <TextBlock/>
    <Button/>
  </StackPanel.Children>
</StackPanel>
```

XAML parser

```
var stackPanel = new StackPanel();
stackPanel.Children.Add(new TextBlock());
stackPanel.Children.Add(new Button());
```

Use Collection Syntax

```
<StackPanel>
  <StackPanel.Children>
    <TextBlock/>
    <Button/>
  </StackPanel.Children>
</StackPanel>
```

```
[ContentProperty(Name = "Children")]
public class Panel : FrameworkElement
{
  public UIElementCollection Children { get ... }
  ...
}
```

Use Collection Syntax

```
<StackPanel>
  <StackPanel.Children>
    <TextBlock/>
    <Button/>
  </StackPanel.Children>
</StackPanel>
```

Use Collection Syntax

```
<StackPanel>

    <TextBlock/>
    <Button/>

</StackPanel>
```

Summary

Elements are mapped to classes

- Except property elements

Attributes are mapped to properties or events

Set properties in XAML

- Attribute syntax
- Property element syntax
- Content syntax
- Collection syntax

Up Next:
Building a User Interface